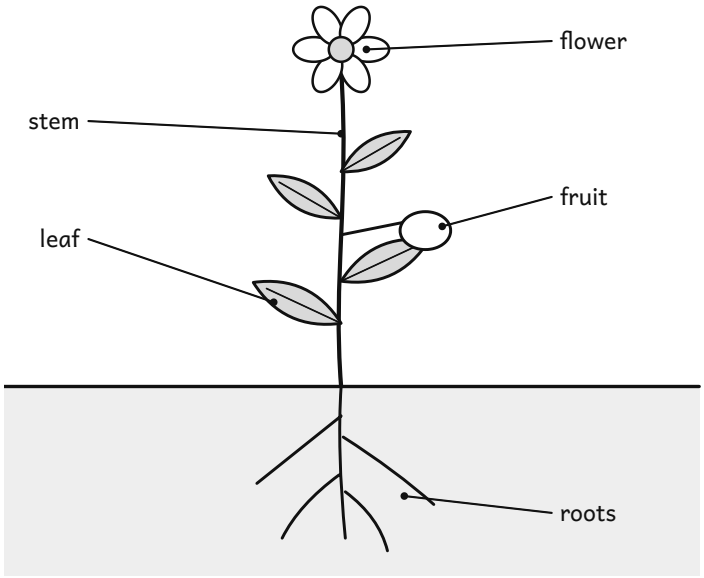
LIVING THINGS

Plants and Their Parts

Roots, stems, leaves and flowers, and what plants need to grow



In this pack

- **Learn it:** Plants and their parts ★☆☆
- **Fact check:** Parts of a plant ★☆☆
- **Think it through:** Why did it happen? ★★☆
- **Investigate:** Does a plant need light to grow? ★★☆
- **Real-world case study:** Mina's rooftop garden ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science: Stage 3 Unit 1 Plants (1.1–1.4).

This pack belongs to: _____ **Class:** _____

© InquiryMind · Original learning material for Grade 2 (PHIS Science continuum, IB PYP Phase 2, and Cambridge Primary Science Stages 2–3) · Illustrations drawn for InquiryMind · Pictures from OpenMoji (openmoji.org), CC BY-SA 4.0 · Print in black and white on A4 · Challenge: ★ core ★★ deeper ★★★ stretch

LEARN IT · A1 · PLANTS

Plants and their parts

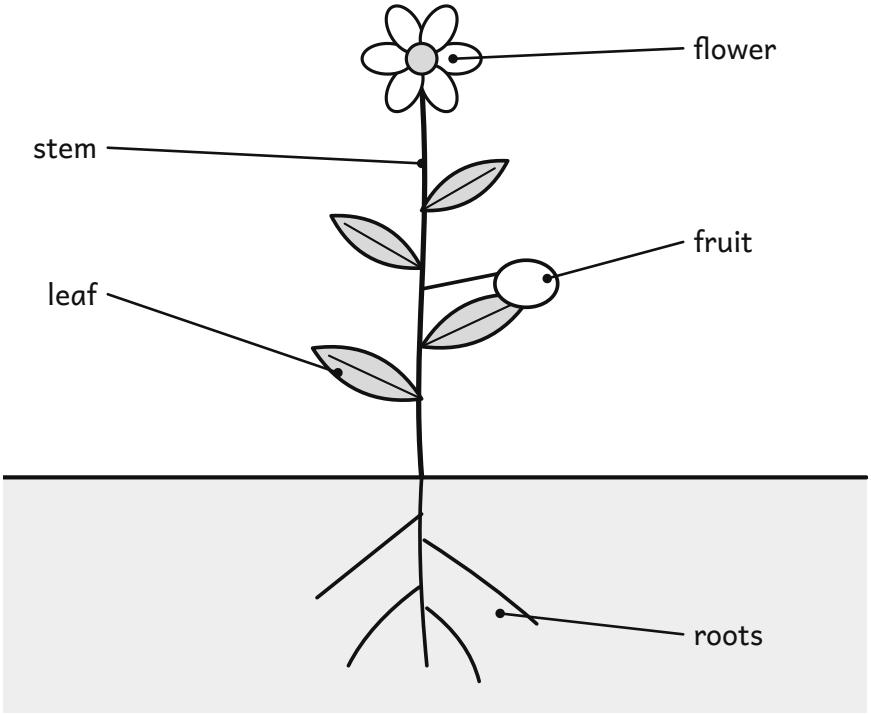
★☆☆
Grade 2

Name: **Class:** **Date:**

Learning goals

- Name the parts of a plant and say what each part does.
- Explain that plants need water, light and the right temperature to grow.
- Explain how water travels from the roots up the stem.

Plants are living things. They grow all around us: rice in the fields, mango trees in the garden, and the shapla flower in the pond. Most plants have the same parts.

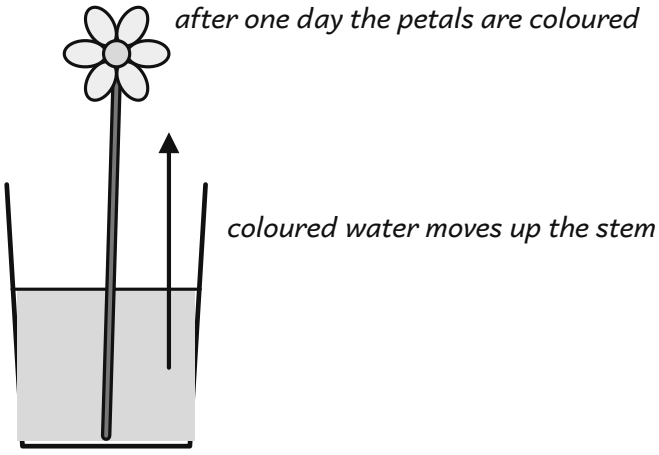


The parts of a plant.

Part	What it does
roots	hold the plant in the soil and take in **water** from the soil
stem	holds the plant up and carries water from the roots to the leaves
leaves	use **sunlight** to make food for the plant
flower	makes seeds, so new plants can grow
fruit	holds and protects the seeds

How water travels

Water goes **in** through the **roots**. Then it travels **up** the **stem**, like water up a straw, to the leaves and flowers.



Put a white flower in coloured water. What happens?

What plants need

water	light	warmth	soil and minerals	air

Plants need **water**, **light**, **air**, the right **temperature** (not too hot, not too cold) and **minerals** from the soil. Without water, a plant **wilts** (droops). Without light, it turns **yellow** and weak. Plants need **healthy** roots, stems and leaves to grow well.



healthy plant



no water: it wilts

Bangladesh connection

Bangladesh is warm and wet, so plants grow very well here. Farmers plant rice when the monsoon rains come.

Try this at home

Plant two bean seeds in cups. Water one every day and do not water the other. What happens after a week?

Key words

roots	take in water from the soil
stem	carries water up the plant
leaves	make food using sunlight
flower	makes seeds
wilt	droop because there is not enough water

FACT CHECK · A1 · PLANTS

Parts of a plant

★☆☆ Grade 2

Name: _____ **Class:** _____ **Date:** _____

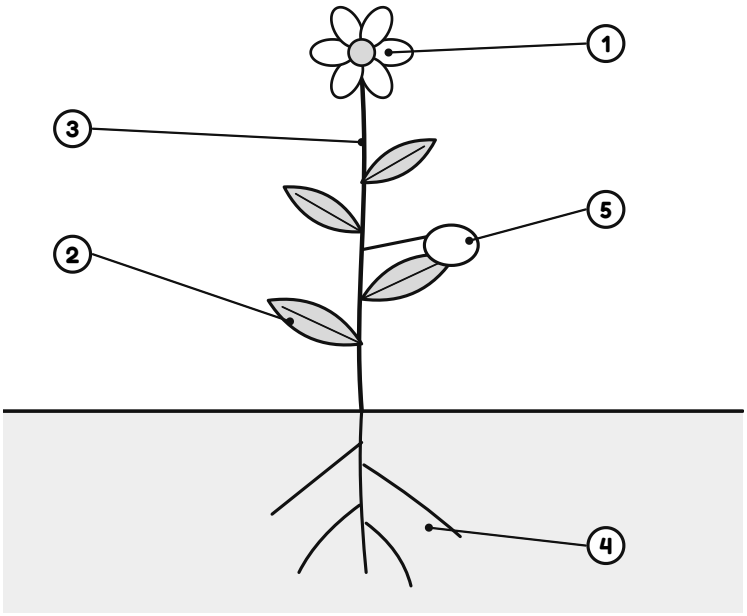
Learning goals

- Label a plant.
- Say what plants need.

1 Label the plant. [5]

LABELS

flower leaf roots stem fruit



1 _____ 2 _____ 3 _____

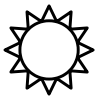
4 _____ 5 _____

2 Draw a line to match. [4]

Draw a line to match each pair, or write the letter.

roots ●	● **A** take in water
stem ●	● **B** make food
leaves ●	● **C** carries water up
flower ●	● **D** makes seeds

3 **Circle what a plant needs to grow.** [3]

 water	 light	 chocolate	 warmth	 a ball

4 **Fill in the gaps.** [3]

WORD BANK

flower **stem** **light** **roots**

Water goes into the plant through the _____. It travels up the _____. Leaves need _____ to make food.

THINK IT THROUGH · A1 · PLANTS

Why did it happen?

★★☆
Grade 2

Name: Class: Date:

Learning goals

- Explain what happens when plants do not get what they need.

1 **Look at the two plants. What happened to plant B? Why?** [2]



A



B

2 **A plant was kept in a dark cupboard for a week. What colour will its leaves be?** [1]

○ dark green ○ yellow

○ red

3 **Why does a plant need healthy roots?** [2]

4 **A plant was left outside on a very cold night. Predict what might happen.** [1]

INVESTIGATE · A1 · PLANTS

Does a plant need light to grow?

★★☆ Grade 2

Name: Class: Date:

Learning goals

- Do a simple fair test.
- Measure and compare.

Class 2B grew two bean plants. Plant A was by the window. Plant B was in a dark cupboard. They both got the same water. After 10 days they measured them.

1 **Complete the table.** [2]

	Plant A (light)	Plant B (dark)
Height (cm)	12	15
Colour of leaves	green	yellow
Healthy?		

2 **Plant B is taller, but is it healthier? Why?** [2]

.....

.....

3 **What did they keep the same?** [1]

.....

4 **Draw the two plants.** [2]

A: in the light

B: in the dark

REAL-WORLD CASE STUDY · A1 · PLANTS

Mina's rooftop garden

★★★ Grade 2

Name: **Class:** **Date:**

Learning goals

- Use plant science in real life.
- Solve a problem.

Science | Food | Maths

Mina lives in a flat in Dhaka. There is no garden, so her family grows vegetables in pots on the **roof**: tomatoes, chillies and spinach (*palong shak*). Mina waters the pots every evening.

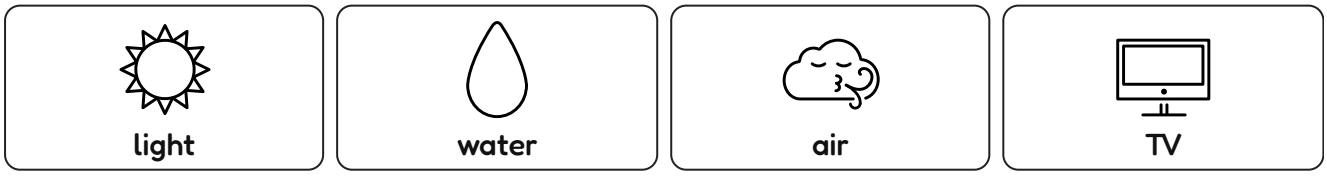
In April the weather is very hot. Some plants start to wilt in the middle of the day. Mina's mother puts up an old cloth to give some shade, and Mina gives them water early in the morning too. Soon the plants are green and strong again.

A story about a rooftop garden in Dhaka.

1 **Why did the plants wilt?** [2]

2 **What did Mina and her mother do to help the plants?** [2]

3 **What do the plants on the roof need? Circle them.** [3]



REVIEW AND CHALLENGE · A1 · PLANTS

Show what you know

★★★
Grade 2

Name: Class: Date:

Learning goals

- Use plant words.
- Answer questions.

1 Find the words.

A	R	C	W	R	M	T	K
D	J	O	F	I	U	W	P
W	Y	J	O	V	V	E	F
A	S	F	S	T	E	M	L
T	W	S	O	E	O	B	O
E	A	F	S	E	E	D	W
R	L	V	L	E	A	F	E
O	O	U	E	A	P	D	R

flower leaf root seed stem water

2 Which part takes in water? [1]

◯ flower ◯ roots

◯ leaf

3 Which part makes seeds? [1]

◯ stem ◯ flower

◯ roots

4 Challenge. Draw a plant and label four parts. [4]

My plant

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name the parts of a plant	○	○	○
say what each part does	○	○	○
say what plants need	○	○	○
explain how water moves up a stem	○	○	○
test if plants need light	○	○	○